Abraham Justin

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Education

Vellore Institute of Technology — Chennai, India
Master of Technology (Integrated) in Computer Science and Engineering – Business Analytics — *Aug. 2023 – May 2028*
CGPA: 9.21/10

Research Experience

Summer Research Intern — May 2025 – July 2025
Vellore Institute of Technology — *Chennai, India*

- Reviewed 20+ papers on quantum computing, quantum algorithms, quantum machine learning, and emerging computing architectures for a faculty-guided literature survey.
- Explored Qiskit simulations and applications in optimization, cryptography, machine learning, and computer architecture; contributed to technical writing and publication planning.

Projects

GitInsight – GitHub Repository Health Platform | *React, Node.js, Express.js, MongoDB, Redis, BullMQ, Gemini*
- Built a full-stack platform that imports a repository's commits, pull requests, issues, reviews, releases and contributors, and scores its health across six weighted dimensions with a confidence level.
- Designed a 13-rule bottleneck detector (e.g. low bus factor, slow first reviews, declining activity) that reports the evidence behind each finding; Gemini writes a structured-JSON assessment grounded only in the computed metrics.
- Engineered incremental, rate-limit-aware GitHub sync as BullMQ jobs on a separate worker, with batched MongoDB bulk upserts, Redis caching and Passport.js authentication, in a ∼5,000-line layered backend.

LexInsight – Explainable Legal Precedent Search | *Python, scikit-learn, BM25, Doc2Vec, FastAPI, React, TypeScript*
- Built a search engine over 781 Indian Supreme Court judgments that classifies query intent (factual, legal issue, statute, supporting or contrasting precedent) with a linear SVM: 0.90 accuracy vs. 0.74 for a keyword baseline.
- Fused five ranking signals (BM25, Doc2Vec, a legal scorer for Acts and provisions, citation treatment and outcome match) with intent-specific weights, showing a score breakdown and explanation for every result.
- Wrote rule-based extractors for Acts, provisions, citations and outcomes (0.89 F1 on provisions) and a FastAPI + React/TypeScript app with search, an evaluation dashboard and a relevance-labelling tool.

Agentic CRM – Lead-Scoring Agent (team of 7) | *Python, FastAPI, scikit-learn, SHAP, MCP*
- Built the lead-scoring agent for a multi-agent CRM: it scores each new enquiry, assigns a Hot, Warm or Cold bucket, and writes back its top three reasons from per-lead TreeSHAP values.
- Audited ∼9,200 real leads for target leakage and removed nine post-contact fields; the calibrated random forest reached **0.87 ROC-AUC** (vs. 0.80 for logistic regression) on 1,841 held-out leads, with Hot leads converting at 80%.
- Integrated through the team's MCP tool server with REST fallback, checksum-verified model loading and retries; covered by 67 tests including a fake MCP server, with 0.25 s median scoring latency.

MammoGraph – Explainable Mammogram Analysis | *Python, PyTorch, Graph Transformer, Grad-CAM, RAG*
- Developed a graph neural transformer on VinDr-Mammo (20,000 images, 10 finding types): ResNet-18 patch features form an 8-neighbour graph, and graph-attention layers predict image-level findings and lesion boxes.
- Raised macro AUROC from 0.64 (ResNet-18 baseline) to **0.82** on a 4,000-image test set; validated Grad-CAM heatmaps with masking checks (peak inside a real lesion 63% of the time vs. 7% for a random point).
- Generated cited three-part summaries using BM25 retrieval over a BI-RADS knowledge base and a local MedGemma model, with a rule-based guard that rejects diagnosis wording, unsupported numbers and unknown citations.

Technical Skills

Languages: Java, Python, JavaScript, TypeScript, C/C++, SQL, HTML/CSS
ML / AI: PyTorch, PyTorch Geometric, TensorFlow, Keras, scikit-learn, XGBoost, SHAP, GNNs, Transformers, RAG, FAISS
LLMs & Agents: Gemini API, Ollama, LangGraph, Model Context Protocol (MCP)
Frameworks: React, Node.js, Express.js, FastAPI, Prisma, Passport.js, Tailwind CSS
Databases: MongoDB, PostgreSQL, Redis, SQLite
Tools: Git, GitHub, BullMQ, Apache Kafka, Apache Spark, pandas, NumPy, Matplotlib, Sentence Transformers, Colab, Kaggle

Achievements

- Secured **1st Rank** in the Computer Science (Business Analytics) branch during the 2023 academic year.
- Ranked among the **Top 10** students in the Computer Science (Business Analytics) branch in 2026.
- Achieved a **1550+ LeetCode Contest Rating** with **150+ problems solved**.